

Homework-2: SparkSQL and SQL on PostgreSQL

Systems and Toolchains for AI Engineers

Deadline: September 18th, 11:59PM ET

In this homework, you will be working with SQL and SparkSQL, specifically with the variant Postgres. You will do your work on a Postgres database and cut-and-paste your answers into a Jupyter Notebook or SQL files (files you create that end with .sql extension).

Submit the HW in **one Jupyter notebook** (or a python file with clear comments). If you need to include scripts outside Jupyter for a given question, note it in your notebook. If you need to include **output** or **screenshots**, you may include them in your notebook or attach them as separate image files using the naming convention qx.png (where x is the question number). If you have more than one image for a given question, name the files as qx_1.png, qx_2.png, etc.

A sample structure of your notebook is shown below

Homework-2 (Sample Example Output to Show Doc Organization)

Q1

Refer to q1.sql and q1.png

Q2

```
from pyspark.sql.functions import monotonically_increasing_id

### Some code here
## For output, refer to q2.png
```

If the output has many rows, it is sufficient to take a screenshot of the first five, unless otherwise instructed.

You will use GitHub classroom to create your GitHub repository. The URL for this assignment's GitHub classroom is: <https://classroom.github.com/a/uJBidJQX>

- Don't create a public repository in your own GitHub account. Use the URL above to create the GitHub repository for this assignment. It includes a starter ReadMe file.
- On Gradescope, only submit a URL to your GitHub repository.
- Not following the submission guidelines may result in grading penalties.

Question 1

(15 points)

Create an empty table in Postgres database to store the Google News records from an RSS Feed. The table should include the following columns:

- lastBuildDate (shared among all entries)
- Title
- Link
- PubDate
- Description
- Source

Here is a sample view of the data points that will need to be stored from the feed into your table.

```
<?xml version="1.0" encoding="UTF-8" ?><rss version="2.0" xmlns:media="http://search.yahoo.com/mrss/"><channel><title>Google News</title><lastBuildDate>Fri, 05 Sep 2025 23:39:10 GMT</lastBuildDate><link>https://news.google.com/</link><description>Google News</description><item><title>Exclusive | How effective are the technology breakthroughs that could extend human life? - New York Post</title><link>https://news.google.com/rss/articles/CBMiswFBVV95cUxPYW5qQTU0ZTFpU1R5T1FrN2pzUGpPU2w3dG5VT19wLWMyQU1lN29Qd1kzWU51bHZqSTRCRXVybWdoZ1c5YzFyYXJWbXhVbmRUOU9XeWtIWEVRbjhkvFhXoc55</link><pubDate>Thu, 04 Sep 2025 21:21:00 GMT</pubDate><description><a href="https://news.google.com/rss/articles/CBMiswFBVV95cUxPYW5qQTU0ZTFpU1R5T1FrN2pzUGpPU2w3dG5VT19wLWMyQU1lN29Qd1kzWU51bHZqSTRCRXVybWdoZ1c5YzFyYXJWbXhVbmRUOU9XeWtIWEVRbjhkvFhXoc55" target="_blank">Exclusive | How effective are the technology breakthroughs that could extend human life?</a>&nbsp;&nbsp;&nbsp;<font color="#666666">New York Post</font></description><source url="https://nypost.com">New York Post</source></item>
```

Design your table with an appropriate unique identifier and select suitable column data types. Your table must be defined within its own DB schema named “news”.

For this question, submit the following:

- .sql file with the script that you used to create the table.
- A screenshot of the column data types displayed using the proper PostgreSQL command to show the table schema and description.
 - The screenshot must include your name. Open a text editor application, write your name, and then take a screenshot of the entire screen including the requested information above. Don't photoshop your name on the image.

Question 2

(15 points)

Populate your table in Q1 with the data in this Google news feed URL:

<https://news.google.com/rss/search?q=technology&hl=en-US&gl=US&ceid=US:en>

There could be large amounts of data in this URL (or a similar one). So, choose an appropriate technique to populate the table in a way that allows scalability.

Use the values in the HTML tag and discard any attributes. For example, in the screenshot provided in Q1, the value of the source field is “New York Post”. Discard “https://nypost.com”.

For this question, submit the following:

- The code you used to populate the table.
- A screenshot of fetching random 5 records from your PostgreSQL DB table. Use PgAdmin to fetch the records. Include the SQL query you used in your screenshot.
 - The screenshot must include your name. Open a text editor application, write your name, and then take a screenshot of the entire screen including the requested information above. Don't photoshop your name on the image.

Question 3

(20 points)

Write a function to find all the news that were published in the last 24 hours. Make sure 1) your function is scalable to run on large amounts of data and 2) your function is dynamic to run on any day (i.e., don't hardcode the date).

For this question, submit the following:

- Your code.
- Screenshot of the output of the function. For this requirement, you may elect to display the output in your notebook or attach it as a separate image.
 - If you attach the output as a separate image, your screenshot must include your name. Open a text editor application, write your name, and then take a screenshot of the entire screen including the requested information above. Don't photoshop your name on the image.

Question 4

(20 points)

You decided to extend your table to include non-technology related news as well. Update your table to add a new column called “category” and pre-populate it with the value of “technology” for existing records. Then, populate the table with the following feeds:

- Business: <https://news.google.com/rss/search?q=business&hl=en-US&gl=US&ceid=US:en>
Use category value of “business” for these news records.
- Sports: <https://news.google.com/rss/search?q=sports&hl=en-US&gl=US&ceid=US:en>
Use category value of “sports” for these news records.

Use a scalable approach to address the requirements in this question.

For this question, submit the following:

- Your SQL script to update the table in a separate SQL file.
- Your code to populate the tables.
- Query all the distinct categories in your notebook. Print the output or attach it as a separate image.
 - If you attach the output as a separate image, your screenshot must include your name. Open a text editor application, write your name, and then take a screenshot of the entire screen including the requested information above. Don't photoshop your name on the image.

Question 5

(15 points)

Delete all the records whose titles include the term “NFL”, case-insensitive. Use a scalable approach that can handle large volumes of records in a large DB table and execute smoothly in real time.

For this question, submit the following:

- Your code.
- Screenshot of the output of a query to select all the records that include the term “NFL”, case-insensitive, after executing your code. Print the output or attach it as a separate image.
 - If you attach the output as a separate image, your screenshot must include your name. Open a text editor application, write your name, and then take a

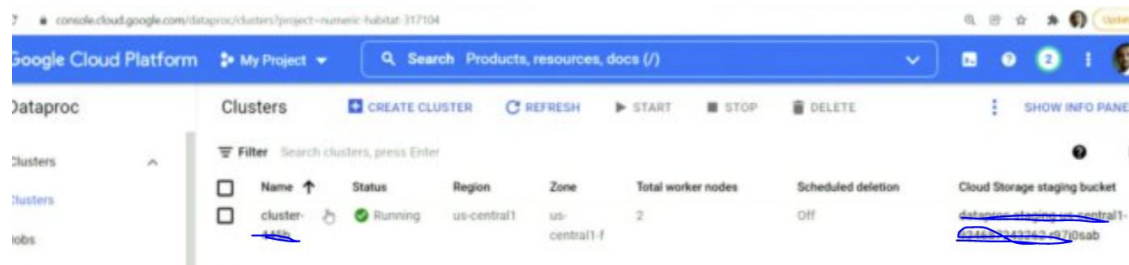
screenshot of the entire screen including the requested information above.
Don't photoshop your name on the image.

Question 6

(15 points)

Create your first cluster on Google Cloud and Submit a screenshot of your cluster while it's "Running". Name your cluster after your Andrew ID (and add any necessary characters).

Your screenshot should look like the following (except for the cluster name):



Don't forget to disable your Google Cloud billing after you are done!